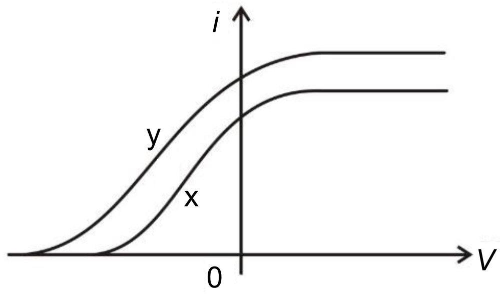
27 The figure shows the variation of the photoelectric current i with voltage V between the electrodes in a photocell for two different radiations, x and y.



The intensity and the frequency of radiation x are I_x and f_x while the intensity and the frequency of radiation y are I_y and f_y. Which of the following shows the relationship between the intensities and frequencies of x and y?

A $I_x > I_y$, $f_x > f_y$

B $I_x > I_y$, $f_x < f_y$

C $I_x < I_y$, $f_x > f_y$

D $I_x < I_y$, $f_x < f_y$

Ans: D

$hf = eV_s + \Phi$,

$f_x < f_y$ since $V_{sx} < V_{sy}$

Since $i_x < i_y$, $I_x < I_y$

28 An electron of mass m and charge e is accelerated from rest through an electric field of potential